ESKWELA Software Testing & Quality Assurance (QA) Suite

Project: ESKWELA Mobile & Web System | **QA Lead:** Software Testing & QA Engineer

Code Repository: https://github.com/jameshofril06-jpg/SkwelaApp

1. Test Plan & QA Strategy

This QA Suite documents the test execution plan, functional test cases, integration API test scripts, User Acceptance Testing (UAT) scenarios, and bug tracking resolution matrix for **ESKWELA**. Testing covers both Student productivity modules and Administrator moderation capabilities.

2. Functional Test Cases

Functional Test Suite (Core System Modules)

TC ID	Module	Test Condition / Step	Expected Result	Status
TC-01	Authentication	Enter valid email and password in Login Activity	User authenticated, redirected to DashboardActivity	**PASSED**
TC-02	Account Guard	Attempt login with suspended/deactivated account	Blocked with restriction notice, logged out	**PASSED**
TC-03	Study Load OCR	Upload study load photo via Camera or Gallery	ML Kit extracts courses and auto-plots to timetable	**PASSED**
TC-04	Schedule Calendar	Switch between Day, Week, and Month views	Filters classes accurately by date range	**PASSED**
TC-05	Time Conflict	Add class overlapping existing class time	Displays ■■ Conflict warning with override option	**PASSED**
TC-06	Schedule Delete	Long press class card and enter class code	Validates input and removes schedule from Firestore	**PASSED**
TC-07	Task Manager	Add new task with priority and course link	Task saved, rendered in list with badge	**PASSED**
TC-08	Task Drop	Tap drop task button and enter reason	Prompts confirmation and records drop reason	**PASSED**
TC-09	AI Quiz Runner	Select study load course and complete 5-15 Qs	Generates questions, tracks score & percentage	**PASSED**
TC-10	Leaderboard	Complete quiz with high score percentage	Score recorded, updates live Leaderboard rankings	**PASSED**
TC-11	Focus Timer	Select focus preset (2m-60m) and start timer	Runs countdown timer, saves session upon completion	**PASSED**
TC-12	Dark Mode	Toggle Dark Mode switch in Profile settings	App appearance instantly flips to dark theme	**PASSED**

3. Integration & API Testing

Integration & REST API Test Suite

TC ID	Module	Test Condition / Step	Expected Result	Status
INT-01	Firebase Auth	Postman POST to identitytoolkit signUp / signIn	Returns HTTP 200 OK with idToken and refreshToken	**PASSED**
INT-02	Firestore Users	GET document /users/student-uid-101	Returns user profile JSON fields (fullName, status)	**PASSED**
INT-03	Firestore Schedule	Run query for schedules where userId == studentUid	Returns JSON array of enrolled classes for date	**PASSED**
INT-04	Firestore Quiz	Run query for ai_quizzes ORDER BY percentage DESC	Returns top 5 leaderboard scores	**PASSED**
INT-05	GitHub OTA API	GET api.github.com/repos/.../releases/latest	Returns HTTP 200 OK with browser_download_url	**PASSED**

4. User Acceptance Testing (UAT) Scripts

User Acceptance Testing (UAT) Scenario Scripts

TC ID	Module	Test Condition / Step	Expected Result	Status
UAT-01	Student Flow	Student logs in ➔ Scans study load photo ➔ Takes AI quiz ➔ Sets Pomodoro timer	End-to-end workflow completed seamlessly	**PASSED**
UAT-02	Admin Flow	Admin logs in ➔ Searches student ➔ Applies 24h suspension ➔ Posts announcement	Suspension countdown rendered, broadcast posted	**PASSED**
UAT-03	OTA Update Flow	User taps 'Check for update (OTA)' in Profile settings	Displays release notes, downloads APK, opens installer	**PASSED**

5. Initial Bug Tracking Log & Resolution Matrix

Bug ID	Description / Defect	Severity	Root Cause Analysis	Resolution / Fix Applied	Status
BUG-01	Scanned study load did not reflect on specific day tabs	HIGH	Schedules were only seeded for specific dates instead of full weekdays.	Updated ScheduleActivity to plot study load across all weekdays.	RESOLVED
BUG-02	OTA update download failed with HTTP 404	HIGH	DownloadManager failed on GitHub's /releases/latest/ 302 redirect.	Switched to GitHub /releases API index 0 and added cache-busting timestamp.	RESOLVED
BUG-03	Text unreadable when editing profile in Dark Mode	MEDIUM	Profile setup layout contained hardcoded dark hex text colors.	Replaced hardcoded hex colors with dynamic @color/text_primary tokens.	RESOLVED
BUG-04	Time schedule overlap allowed duplicate class booking	MEDIUM	AddClassActivity lacked start/end time overlap validation.	Implemented parseTimeToMinutes() and overlap check with warning popup.	RESOLVED
BUG-05	Task deletion occurred accidentally on single tap	LOW	No confirmation dialog before removing task records.	Added AlertDialog requiring confirmation and drop reason prompt.	RESOLVED